

Emergency Food Network (EFN) Service Data

Data Structure, Definitions, and Validation

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```
In [1]: import pandas as pd
import numpy as np

df = pd.read_excel("2014-2024_meals_no_partners.xlsx")
```

1. Data Generating Process (DGP)

Unit of Observation and Scope

Each observation represents a service unit–calendar year record of service activity associated with Emergency Food Network (EFN) partner organizations. The service unit identifier is recorded in a field labeled zip and includes both standard five-digit ZIP codes and non-geographic program identifiers corresponding to EFN-operated or partner programs (e.g., home delivery programs, medically tailored food initiatives, or time-limited relief efforts).

The dataset therefore forms an unbalanced panel of service units over time. Service units—whether geographic ZIP codes or program-level aggregations—may enter or exit the dataset across years due to changes in partner participation, program availability, service coverage, or reporting practices.

The data capture service activity rather than population counts. Observations reflect individuals who received food assistance through EFN partners or programs and do not represent the full resident population of a ZIP code nor the full eligible population for any given program. Geographic analyses in subsequent sections explicitly distinguish ZIP-coded observations from program-level observations to avoid conflating geographic and programmatic service patterns.

Key Variables Used in the Analysis

The table below summarizes the primary variables used throughout the analysis.

```
In [2]: variable_definitions = pd.DataFrame({
    "variable": [
        "zip",
        "year",
        "total_undup",
        "total_dup",
        "total_clients",
        "efn_pounds_to_partner",
        "pounds_dist_partner"
    ],
    "definition": [
        "Service unit identifier; includes both five-digit ZIP-coded u",
        "Calendar year of recorded service activity.",
        "Number of unique individuals served at least once within the",
        "Number of repeat service visits beyond the first visit within",
        "Total recorded service encounters within the year; defined as",
        "Pounds of food distributed by EFN to partner organizations as",
        "Pounds of food distributed by partner organizations to client"
    ]
})

variable_definitions_display = variable_definitions.copy()

variable_definitions_display.style.hide(axis="index").set_properties(*
    "text-align": "left",
    "white-space": "normal"
).set_table_styles([
    {"selector": "th", "props": [("text-align", "left")]},
    {"selector": "td", "props": [("padding", "6px")]}
])
```

variable	definition
zip	Service unit identifier; includes both five-digit ZIP-coded units and program-level service identifiers.
year	Calendar year of recorded service activity.
total_undup	Number of unique individuals served at least once within the year.
total_dup	Number of repeat service visits beyond the first visit within the year.
total_clients	Total recorded service encounters within the year; defined as total_undup + total_dup.
efn_pounds_to_partner	Pounds of food distributed by EFN to partner organizations associated with a service unit.
pounds_dist_partner	Pounds of food distributed by partner organizations to clients associated with a service unit.

Client Counts and Visit Structure

Client activity is recorded using unduplicated and duplicated measures.

Unduplicated counts represent the number of unique individuals who received food assistance at least once during the calendar year. Each individual contributes exactly one count to the unduplicated total in a given year.

Duplicated counts represent additional service visits beyond the first visit per individual during the same year. These counts capture repeat utilization and therefore measure service intensity rather than the number of people.

Total service encounters (visits) are defined as the sum of unduplicated and duplicated counts:

$$\text{Total service encounters} = \text{Unduplicated} + \text{Duplicated}$$

This identity holds exactly for service unit-year observations, as verified in Section 2.

Age Classification

Clients are classified into mutually exclusive age categories:

- 0–2 years
- 3–18 years
- 19–54 years
- 55 years and older
- Unknown age

Age-specific counts are recorded separately for unduplicated individuals and duplicated visits.

Food Distribution Measures

Two aggregate measures of food volume are recorded:

- **EFN pounds to partners**, representing pounds of food distributed by EFN to partner organizations associated with a given service unit.
- **Pounds distributed by partners**, representing pounds of food distributed by partner organizations to clients associated with a given service unit.

For ZIP-coded service units, these measures reflect food flow within a geographically defined area. For program-level service units, they reflect food distribution aggregated at the program level rather than within a specific ZIP code.

These measures capture distinct stages of food flow within the EFN distribution system and are treated as indicators of service capacity and delivery.

Temporal and Geographic Structure

The dataset spans multiple calendar years and includes variation in service volume and service unit coverage over time. Both ZIP-coded service units and program-level service units may enter or exit the dataset as partner operations or programs begin, end, or change reporting practices.

Missing ZIP-year observations are interpreted as reflecting changes in service coverage, program structure, or reporting rather than recorded zero service activity.

Intended Use and Analytical Limits

This dataset supports descriptive analysis of service reach and utilization, demographic composition, repeat service usage, and service capacity within observed service units. Geographic analyses are restricted to ZIP-coded service units, while program-level service units are analyzed separately to avoid conflating geographic and programmatic service patterns.

What the data measure: This dataset measures recorded service activity within the Emergency Food Network (EFN) partner system at the service unit-year level. It captures annual counts of unique clients served, repeat service visits, and selected food distribution volumes associated with observed service units.

What the data do not measure: These records should not be interpreted as direct measures of underlying food insecurity in the broader community, the full eligible population within a ZIP code, or unmet need outside the observed EFN network. Because the unit of observation is a service unit-year record, the dataset describes service activity within observed units rather than countywide person-level outcomes.

2. Accounting Identity Validation

Internal consistency checks were performed to verify that:

- Total service encounters equal the sum of unduplicated and duplicated counts for all ZIP–year observations.
- Age-specific unduplicated counts sum exactly to total unduplicated counts.
- Age-specific duplicated counts sum exactly to total duplicated counts.

All accounting identities hold exactly across the dataset.

3. Zeros and Missingness

This section examines the presence and structure of zero values in client counts and food distribution measures across all service unit–year observations, including both ZIP-coded and program-level service units. The goal is to distinguish plausible zero service activity from potential reporting gaps or structural features of the data before proceeding to descriptive analysis.

```
In [3]: zero_summary = pd.DataFrame({
    "variable": [
        "total_clients",
        "total_undup",
        "total_dup",
        "pounds_dist_partner",
        "efn_pounds_to_partner"
    ],
    "number_of_zeros": [
        (df["total_clients"] == 0).sum(),
        (df["total_undup"] == 0).sum(),
        (df["total_dup"] == 0).sum(),
        (df["pounds_dist_partner"] == 0).sum(),
        (df["efn_pounds_to_partner"] == 0).sum()
    ]
})

zero_summary
```

```
Out[3]:
```

	variable	number_of_zeros
0	total_clients	0
1	total_undup	0
2	total_dup	2
3	pounds_dist_partner	0
4	efn_pounds_to_partner	1

```
In [4]: combo_checks = {
    "clients_gt0_pounds_dist_eq0": (
        (df["total_clients"] > 0) & (df["pounds_dist_partner"] == 0)
    ).sum(),

    "clients_eq0_pounds_dist_gt0": (
        (df["total_clients"] == 0) & (df["pounds_dist_partner"] > 0)
    ).sum(),

    "clients_gt0_efn_pounds_eq0": (
        (df["total_clients"] > 0) & (df["efn_pounds_to_partner"] == 0)
    ).sum()
}

combo_checks
```

```
Out[4]: {'clients_gt0_pounds_dist_eq0': 0,
        'clients_eq0_pounds_dist_gt0': 0,
        'clients_gt0_efn_pounds_eq0': 1}
```

Summary of Zero-Value Diagnostics

Zero-value diagnostics indicate a high degree of internal consistency in the data. No service unit–year observations report zero total clients, and no cases occur in which food distribution is reported without corresponding client counts or vice versa.

Zero values appear only in limited component measures. Specifically, two service unit–year observations report no duplicated visits, indicating that all clients in those cases received food exactly once. In addition, one service unit–year observation reports zero EFN pounds delivered to partners while client service and partner-level distribution remain positive, consistent with timing differences or inventory carryover.

Overall, zero values observed in the dataset are rare, plausible given the data-generating process, and do not indicate reporting errors or structural inconsistencies.